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**Homework 1: Markov Chains Smart Home Energy Management**

1. <u>State Representation</u>

1.1. The current model does not strictly satisfy the Markov property. This states that the future state depends only on the current state in a sequence of events that preceded it. In this case the exception relies on the definition of the states $S_2$ and $S_3$. Since the problem wants us to analyse the flows of battery in a solar panel, these states represent the exact same state. This means it should not matter if it is Cloudy or Night time, because the state is always low and there is no solar production. Since the probabilities of $S_2$->$S_4$ and $S_3$->$S_4$, which should represent the exact same flow in this problem, are different, it implies the system needs to have context about the time of day, to make the transition to empty.

In order to make this a Markovian chain we should merge states $S_2$ and $S_3$ into a unique "No Sun/Low Battery" state. This eliminates the need for more context when making the decision and redundant behaviour.

1.2. To represent the situation with the old charger, we must expand our state space by adding a fifth state, $S_5$ (Failure). Because the problem states the battery will never charge again once it fails, $S_5$ becomes an absorbing state.

When the system is in $S_4$ (Night/Empty) , there is now a high probability of transitioning to $S_5$ instead of returning to other states. Once the system enters $S_5$, the probability of staying there is $1$ ($100\%$).

To illustrate this, here is an example of the new $5 \times 5$ transition matrix:

$$P_{new} = \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 & 0.0 \\ 0.1 & 0.4 & 0.4 & 0.1 & 0.0 \\ 0.0 & 0.1 & 0.0 & 0.1 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

The first three rows remain similar to the original system, but with a $0.0$ probability of jumping straight to failure. In the fourth row representing $S_4$, we redistributed the original probabilities so that there is now an $80\%$ $(0.8)$ chance of moving to the failure state $S_5$. The fifth row represents our new absorbing state, with a $1.0$ probability on the diagonal, meaning the system can never leave $S_5$.

1.3. To represent a situation where the battery charges slower and never recovers its standard time, we cannot just use a single absorbing state like we did for a complete failure. Instead, we need an entirely new, parallel set of states representing the "Degraded" battery mode.

**Normal States:** $S1$ (Sunny/Full), $S2$ (Cloudy/Low), $S3$ (Night/Low), $S4$ (Night/Empty).

**Degraded States:** $S5$ (Sunny/Full/Slow), $S6$ (Cloudy/Low/Slow), $S7$ (Night/Low/Slow), $S8$ (Night/Empty/Slow).

**Transition Logic:**

1. When in $S4$ (Night/Empty), there is a probability of the battery permanently degrading, meaning it transitions into the new Degraded state space (e.g., jumping from $S4$ to $S8$).
2. Because the battery "never recovers", the probability of transitioning from any Degraded state ($S5$ through $S8$) back to any Normal state ( $S1$ through $S4$) is exactly 0.
3. Within the Degraded states, the probabilities of reaching the "Full" state ($S5$) are significantly reduced compared to the normal states, representing the slower charge.

$$\begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.1 & 0.4 & 0.4 & 0.1 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.1 & 0.1 & 0.1 & 0.0 & 0.0 & 0.0 & 0.7 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.2 & 0.5 & 0.3 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.1 & 0.6 & 0.2 & 0.1 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.05 & 0.45 & 0.4 & 0.1 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

2. <u>State Distribution</u>

2.1. Calculate the state distribution at $t = 2$

At $t = 0$, the house starts in state $S_1$ (Sunny/Full).

- **Initial Probability Vector ($v_0$):**
  $v_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$
- **Transition Matrix ($P$):**
  $$P = \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

The distribution at $t = 1$ ($v_1$) is found by multiplying the initial vector by the transition matrix:

$$v_1 = v_0 P$$

$$v_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0.0 \end{bmatrix}$$

To find the distribution at $t = 2$ ($v_2$), we multiply $v_1$ by the matrix $P$ again:

$$v_2 = v_1 P = \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0 \end{bmatrix} \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0.24 & 0.34 & 0.36 & 0.06 \end{bmatrix}$$

2.2. To find the probability of being in state $S_4$ at $t = 10$, we calculate the state distribution vector $v_{10}$ using the formula:

$$v_{10} = v_0 \cdot P^{10}$$

Given the initial state $v_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$ and the transition matrix $P$:

$$P = \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

In the previous question, we saw that, in order to calculate $v_2$, we simply did computed the product of $v_0$ by the 2nd power of the matrix $P$. Hence, by computing the 10th power of the matrix $P$, the state distribution vector at $t = 10$ is approximately:

$$v_{10} \approx \begin{bmatrix} 0.1872 & 0.3999 & 0.2831 & 0.1298 \end{bmatrix}$$

The probability that the system is in the Blackout state ($S_4$) at $t = 10$ is the fourth component of $v_{10}$:

$$P(X_{10} = S_4) \approx 0.1298$$

This represents a 12.98% probability of blackout at $t = 10$.

2.3. A stationary distribution $\pi$ exists if the Markov Chain is irreducible and aperiodic. It is found by solving the equation $\pi P = \pi$, subject to the constraint $\sum \pi_i = 1$.

The initial transition matrix $P$ is:

$$P = \begin{bmatrix} 0.4 & 0.2 & 0.4 & 0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

To find $\pi$, we solve the system of linear equations:

- $0.4\pi_1 + 0.2\pi_2 + 0.1\pi_3 = \pi_1$
- $0.2\pi_1 + 0.5\pi_2 + 0.4\pi_3 + 0.5\pi_4 = \pi_2$
- $0.4\pi_1 + 0.2\pi_2 + 0.4\pi_3 + 0.1\pi_4 = \pi_3$
- $0.1\pi_2 + 0.1\pi_3 + 0.4\pi_4 = \pi_4$
- $\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$

Solving it gives:

$$\pi \approx \begin{bmatrix} 0.1875 & 0.4000 & 0.2812 & 0.1312 \end{bmatrix}$$

In the case of **Ex 1.2**, state $S_4$ becomes an absorbing state because the charger fails permanently. The transition matrix is modified such that the fourth

row is $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$. In any finite Markov Chain with an absorbing state that is reachable from all other states, the stationary distribution is:

$$\pi = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

Considering **Ex 1.3** and matrix $P$, the normal states ($S_1$ to $S_4$) are transient because there is a non-zero probability of transitioning from $S_4$ to the degraded state $S_8$, and zero probability of returning from the degraded states to the normal states. This means the stationary distribution will be concentrated entirely in the degraded state space.

Solving for $\pi$ requires finding the stationary distribution of the $4 \times 4$ degraded sub-matrix ($P'$):

$$P' = \begin{bmatrix} 0.2 & 0.5 & 0.3 & 0.0 \\ 0.1 & 0.6 & 0.2 & 0.1 \\ 0.05 & 0.45 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

The final stationary distribution for the 8-state system is:

$$\pi \approx \begin{bmatrix} 0 & 0 & 0 & 0 & 0.059 & 0.529 & 0.294 & 0.118 \end{bmatrix}$$